

Holger Pirk

H.Pirk Nutrition Report and recommendations

Prepared by
Session date

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Nutrition Report & Recommendations

Client: Holger Pirk
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Patient Account Number: HPIRK916471488
Consultation Type: Insurance Initial Consultation
Date: 24 March 2026

Reason for Consultation

Holger presented for assessment of fatigue, poor recovery, mood changes, and suboptimal marathon performance despite a high-quality diet and meticulous nutrition tracking. His main goals are to:

- assess whether his current intake is adequate for his training load
- improve recovery and day-to-day energy
- support marathon performance, with a target of **sub-2:57** this season
- ensure nutrition is no longer a limiter to performance

A key concern raised in consultation was whether his current presentation reflects **under-fuelling rather than overtraining**.

Background Summary

Holger has a long running background and has been training seriously over the last year. During this period, he gradually reduced body weight from approximately **90 kg to 79–80 kg** over 7–8 months. He ran a marathon in October 2025 and, despite feeling better prepared for his most recent marathon 4 weeks ago, ran approximately 2 minutes slower than expected.

He reports:

- fatigue without a clear pattern during the day
- poor recovery, including after easy runs
- increased irritability / low mood
- frequent urination, particularly overnight, which improved when carbohydrate intake increased 2 weeks ago
- sleep quality is acceptable, but sleep duration is low at around **6 hours/night**

- no injury history
- no known medical conditions
- no current medication
- vegetarian diet including eggs and dairy

He has not worked with a dietitian previously.

Anthropometry

- **Height:** 198 cm
- **Weight:** 79 kg
- **Reported body fat via smart scale:** reduced from ~13–14% to ~6–7%
 - this is likely an under-estimate, but the downward trend is clinically relevant

Interpretation

Holger is currently very light for his height and training volume. The trend of progressive weight loss, reduced body fat, fatigue, slower recovery, and worsening performance strongly suggests that current energy intake has likely not kept pace with training demands over time.

Training Load

Weekly Training Structure

- **Monday:** 10 km easy
- **Tuesday:** 10 km easy
- **Wednesday:** 12 km speed session
- **Thursday:** 10 km easy
- **Friday:** Rest
- **Saturday:** 12 km easy / parkrun
- **Sunday:** 30–35 km long run
- **Daily:** 20–25 min bodyweight / cross-training

Estimated Weekly Load

- Running volume: approximately **84–89 km/week**
- Estimated total training time: approximately **9–12 hours/week**

Reported Daily Energy Expenditure

- Easy 10 km day: **~3800 kcal**
 - Wednesday speed day: **~4000 kcal**
 - Long run day: **~4700 kcal**
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Current Dietary Pattern

Holger follows a vegetarian diet and eats a wide variety of foods. Food quality appears high, with a strong intake of whole grains, legumes, vegetables, eggs, dairy, tofu, tempeh, seitan, and minimally processed meals.

Typical Breakfast

- 2 eggs
- 200 g chopped tomatoes
- 100 g spinach or 200 g sauerkraut
- 300 g whole groat oat porridge
 - made with 60 g groats + 240 g milk

Estimated intake: **~700 kcal**

Typical Lunch / Dinner

Generally large, whole-food based meals such as:

- beetroot soup with barley and tempeh
- shakshuka with mashed potatoes, peas, Brussels sprouts
- roasted squash with tofu
- Thai coconut / silken tofu curry
- Brussels sprouts with barley and tofu

Dinner intake tends to be substantial, with the majority of calories often “made up” later in the day.

Snacks

- quark
- dates before evening training
- leftover porridge

Fluids / Other

- 2 flat whites daily
- no alcohol
- water intake not tracked, but perceived as adequate

Current Fuelling Practices

Before Training

- porridge most commonly
- waffles or pancakes before races / long runs
- occasional homemade cereal bars

During Training

- long runs: approximately **1 OTE bar/hour (~250 kcal)**

- races: approximately **1 OTE chew every 25 minutes (~100 kcal)**

Average long-run intake appears to be around **30 g carbohydrate/hour**.

After Training

- silken tofu blended with fruit
- leftover porridge with whey protein

Clinical Nutrition Assessment

1. Energy Availability

Holger has been trying to “match” intake to expenditure, but in practice he reports that prior to the last 2 weeks he was often in:

- **~2000 kcal deficit on long run days**
- **~400–450 kcal average daily deficit**

This is highly significant.

Interpretation

This is the central issue in the case.

Holger’s symptoms are highly consistent with **chronic low energy availability**, including:

- fatigue
- poor recovery
- low mood / irritability
- slower-than-expected performance despite appropriate training
- progressive loss of body mass
- possible low carbohydrate availability / nutritional ketosis
- nocturnal urination which improved after increasing carbohydrate intake

This is not a case of poor food quality. It is a case of **insufficient total energy and carbohydrate availability relative to training load**.

2. Carbohydrate Availability

Holger appears to have been relatively low in carbohydrate availability for the demands of marathon training. His symptoms, intake pattern, and recent improvement in urination with increased carbohydrate intake support this.

For an athlete running 84–89 km per week with weekly speed work and long runs, carbohydrate intake is likely the most important limiting factor in current performance and recovery.

Why this matters

Inadequate carbohydrate availability can contribute to:

- reduced glycogen restoration

- slower recovery between sessions
- higher perceived exertion
- poorer quality in speed sessions
- reduced immune and hormonal resilience
- low mood / irritability
- poorer sleep and higher physiological stress
- reduced adaptation to training

3. Protein Intake

Holger reports targeting **130 g/day**, but often eating closer to **180 g/day**.

At 79 kg, evidence-based endurance protein targets are approximately:

- **1.6–1.8 g/kg/day**
- equivalent to **125–142 g/day**

Interpretation

Protein intake is more than adequate and at times likely unnecessarily high. Excess protein is not the main issue here and may be displacing carbohydrate that is more urgently needed for performance and recovery.

4. Fat Intake

Fat makes up the remainder of intake after protein and carbohydrate. Fat is important for:

- endocrine function
- energy density
- satiety
- absorption of fat-soluble vitamins

However, in this case, the primary need is not to push fat higher, but to ensure carbohydrate intake is sufficiently prioritised.

5. RED-S / Low Energy Availability Risk

Holger's presentation is consistent with **moderate RED-S risk / clinically significant low energy availability**, based on:

- high training volume
- prolonged energy deficit
- fatigue and poor recovery
- mood changes
- progressive weight / body fat reduction
- likely low carbohydrate availability
- rigid intake-versus-output mindset

There is no history of stress fracture or known endocrine disturbance reported, which is reassuring, but the current symptom profile still warrants intervention.

6. Recovery & Sleep

Holger reports sleep quality as acceptable, but duration is around **6 hours per night**, which is below ideal for his training load and recovery needs.

Interpretation

Under-fuelling plus inadequate sleep is likely amplifying:

- fatigue
 - slower recovery
 - irritability
 - poorer adaptation to training
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Main Nutrition Priorities

1. **Restore adequate energy availability**
 2. **Increase carbohydrate availability daily and around training**
 3. **Reduce large day-to-day deficits**
 4. **Improve long-run and quality-session fuelling**
 5. **Support recovery with structured post-session intake**
 6. **Reduce over-reliance on rigid intake/output matching**
 7. **Monitor body composition without pursuing further loss**
 8. **Screen for blood markers relevant to low energy availability**
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Nutrition Recommendations

1. Energy Intake Strategy

Immediate phase: next 2–4 weeks

Aim for:

- **maintenance intake at minimum**
- no intentional deficits
- particularly avoid deficits on long-run and quality-session days

Thereafter

If recovery, mood, and performance do not improve sufficiently, progress toward:

- **small surplus of ~150–300 kcal/day**, especially on harder training days

Clinical note

Holger has already increased intake over the last 2 weeks and weight has maintained. This is a positive sign. Maintenance is the correct short-term target. A small surplus may be required thereafter if recovery remains poor or if weight / energy remains suppressed.

2. Carbohydrate Targets

At **79 kg**, recommended carbohydrate intake:

Easy / lower load days

- **5–6 g/kg/day**
- **395–475 g carbohydrate/day**

Moderate / quality training days

- **6–8 g/kg/day**
- **475–630 g carbohydrate/day**

Long run / highest load days

- **7–9 g/kg/day**
- **550–710 g carbohydrate/day**

These are broad targets and should be adjusted based on appetite, GI tolerance, body mass trend, and training phase.

Key practical point

Holger does not need to hit the high end every day immediately. The most important change is to **meaningfully increase carbohydrate availability on hard days and around sessions**, rather than staying relatively flat and under-fuelled across the week.

3. Protein Targets

Aim for:

- **1.6–1.8 g/kg/day**
- **125–142 g/day**

This is sufficient. There is no need to push daily protein much above this routinely.

Distribution target

Aim for **25–35 g protein** across 4–5 eating occasions daily.

4. Fat Targets

Aim for approximately:

- **1.0–1.2 g/kg/day**
- **79–95 g/day**

Fat intake should remain adequate, but should not crowd out carbohydrate intake, particularly before and after key sessions.

5. Pre-Training Fuelling

For morning / key sessions, fuelling should be more intentional.

Easy runs under ~60 minutes

If tolerated, a small carbohydrate top-up may still help if recovery is poor:

- banana
- toast with jam or honey
- small oat-based snack
- dates

Quality sessions / long runs

Aim for:

- **1–3 g carbohydrate/kg in the 1–4 hours pre-session**, depending on timing and GI comfort

Examples:

- porridge + banana + honey
- pancakes / waffles + maple syrup + fruit
- white toast / bagels + jam
- porridge + defatted peanut powder + fruit

If training early and only a small pre-run option is tolerated:

- banana + sports drink
 - toast + honey
 - cereal bar + carbohydrate drink
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6. During Training Fuelling

This is one of the biggest opportunities for improvement.

Current intake

- ~30 g carbohydrate/hour on long runs

Target

Progress toward:

- **60–80 g carbohydrate/hour** during long runs and race-specific work

This should be built gradually in training.

Practical options

- carbohydrate drink
- energy chews
- gels
- bars, if tolerated
- combination approach

Example progression

- week 1: 40 g/hr
- week 2: 50 g/hr
- week 3: 60 g/hr
- then progress toward 70–80 g/hr as tolerated

Recommendation

Use a **carbohydrate-electrolyte drink** during longer runs to make intake easier and reduce reliance on solids alone.

7. Hydration & Sodium

Fluids

Aim for:

- **~2.6 L/day baseline**
- **plus 500–750 mL/hour of training**

This should be adjusted for sweat rate, climate, and session duration.

Sodium / electrolytes

For longer runs or higher sweat loss:

- **1000–1500 mg sodium** across the session / around training as indicated

This can come from:

- electrolyte tablets / powders
 - sports drink
 - salty recovery foods such as **salted pretzels post-session**
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8. Post-Training Recovery

Recovery needs to be made more deliberate and consistent, especially after Wednesday, Thursday, and Sunday sessions.

Immediate target

Within 30–60 minutes of key sessions:

- **1.0–1.2 g carbohydrate/kg**
- **0.3 g protein/kg**

For Holger, this is approximately:

- **80–95 g carbohydrate**
- **20–25 g protein**

Practical examples

- porridge + whey + banana + honey
- recovery smoothie with fruit + oats + whey / silken tofu
- bagels / toast + yoghurt / quark
- cereal + milk + fruit + protein addition
- salted pretzels + recovery drink + fruit

The current post-run intake is a good base, but likely too small for the level of depletion on harder sessions.

9. Meal Timing Structure

A more consistent intake pattern through the day is recommended to reduce energy dips and improve recovery.

Key goals

- avoid long gaps without intake
- avoid leaving too many calories until evening
- fuel earlier around training rather than “making up” later

Suggested structure

- breakfast
- mid-morning snack
- lunch
- pre-training snack
- post-training recovery
- dinner
- optional pre-bed snack

A pre-bed snack can also support recovery, especially on high-load days.

Examples:

- porridge + milk
- yoghurt / quark + fruit + cereal
- toast + nut butter + honey

- cereal + milk

10. Micronutrients / Supplement Recommendations

Core supplements

1. Vitamin D

- **2000 IU/day**
- take with or after food

2. Sports nutrition support

- whey protein isolate
- defatted peanut powder in porridge
- energy chews / oat bars
- electrolytes

3. Omega-3

Increase plant omega-3 sources regularly:

- chia seeds
- ground flaxseed
- walnuts
- hemp seeds
- rapeseed oil
- edamame / soy
- leafy greens

Given vegetarian status and training load, consider an **algal oil EPA/DHA supplement** if intake is low.

4. Magnesium

Previously used. May be reasonable to reintroduce if intake is low or if sleep / muscle relaxation is an issue, though it is not a primary intervention here.

11. Blood Tests Recommended

Given symptoms suggestive of low energy availability, Holger should arrange blood tests with GP / medical team.

Recommended markers to request:

- Full blood count
- Ferritin
- B12
- Folate
- Vitamin D

- Thyroid profile: **TSH, free T4, free T3**
- Testosterone
- LH
- FSH
- Urea & electrolytes
- Liver function
- HbA1c / fasting glucose if clinically indicated
- CRP if needed clinically

These will help assess for iron deficiency, thyroid disturbance, endocrine suppression, and broader consequences of prolonged under-fuelling.

12. Foods to Prioritise More Often

Given his preferences, ideal foods to build around include:

Protein

- tofu
- tempeh
- seitan
- eggs
- quark / yoghurt
- whey protein
- lentils
- defatted peanut powder

Carbohydrate

- oats
- rye
- barley
- buckwheat
- quinoa
- bulghar wheat
- potatoes
- sweet potatoes
- fruit
- bread / wraps / bagels
- rice where needed for lower-fibre fuelling around harder sessions

Easy-to-digest training carbohydrates

- pancakes / waffles
- toast + jam / honey

- bananas
- rice-based snacks
- cereal bars
- sports drinks / chews / gels

Clinical Impression

Holger's overall diet quality is strong, but his nutrition strategy has likely been **too energy-restricted and too conservative in carbohydrate provision** for the demands of marathon training.

The most likely explanation for current symptoms is:

- **chronic under-fuelling**
- **low carbohydrate availability**
- **reduced energy availability / RED-S risk**
- compounded by **suboptimal sleep duration**

This appears to be a performance nutrition issue rather than a motivation or discipline issue. He is doing many things well, but the numbers have been too low for too long.

Key Recommendations Summary

1. **No intentional deficit for the next 2–4 weeks**
2. **Aim for maintenance intake minimum; consider small surplus thereafter if needed**
3. **Increase carbohydrate intake substantially, especially on hard and long-run days**
4. **Work toward 60–80 g carbohydrate/hour during long runs**
5. **Use a carbohydrate-electrolyte drink during longer sessions**
6. **Post-session target: 80–95 g carbs + 20–25 g protein**
7. **Do not over-prioritise protein above needs**
8. **Avoid leaving too much intake until evening**
9. **Aim for 2.6 L fluids/day + 500–750 mL per hour of training**
10. **Include sodium/electrolytes regularly around long sessions**
11. **Complete blood tests**
12. **Review in 2–3 weeks with symptom response, weight trend, recovery, and training quality**

Follow-Up Focus

At next review, assess:

- recovery after easy and quality sessions
- overnight urination
- mood and irritability
- body weight and body composition trend
- training quality and perceived effort
- carbohydrate intake on easy vs hard days

- ability to tolerate 60–80 g/hr during long runs
- blood test results if completed